

Parameterized Restless Temporal Path

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FCT 2025 - University of Wrocław, Poland

Plan

1 Introduction

2 Results

- Uniform delay one (strict setting)
- With arbitrary (positive) delays
- Interval model

3 Conclusion

Temporal graphs: motivations, history

Motivations:

- Graphs represent relationships between entities
- In many real-world applications, those relationships are time-related:
 - Public Transit network: there is a bus that goes directly from stop s to stop t only at time $\tau_1, \tau_2, \tau_3 \dots$
 - Phone call network: Alice calls Bob from time τ to time τ'
 - Social network: Alice follows Bob at time τ
 - ...

History:

- Fragmented literature, plenty of models
- Time-dependent network (e.g.: road networks [Dreyfus 1969])
- Evolving graph (e.g.: ad-hoc networks [Xuan et al. 2003])
- Temporal network (e.g.: social network [Holme et al. 2011]), link stream [Latapy et al. 2017]
- Time-varying graph (ex: distributed computing [Casteigts et al. 2012]), temporal graph (ex: graph theory [Michail 2015])

Temporal graphs: motivations, history

Motivations:

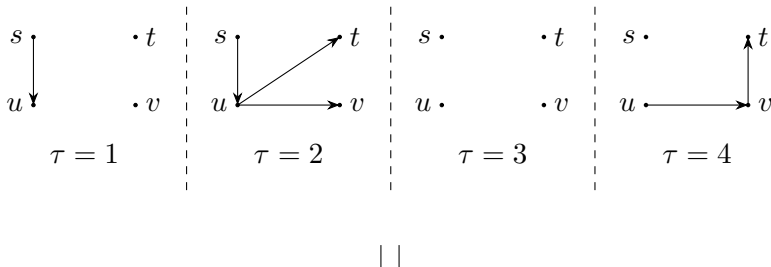
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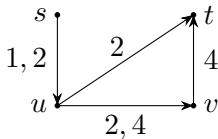
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Temporal graphs: models

Sequence of snapshots:

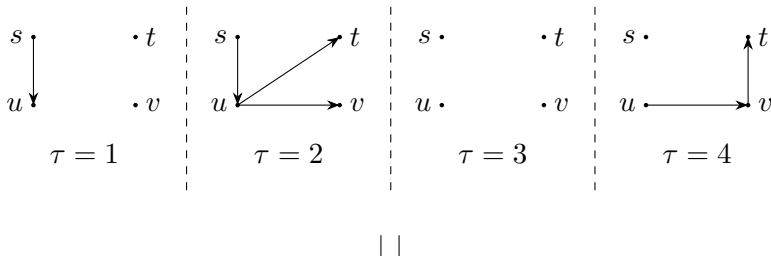


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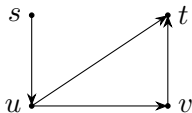


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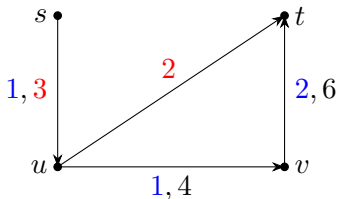
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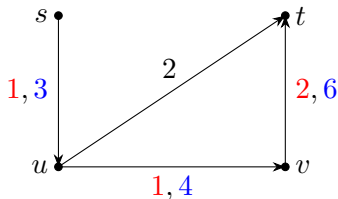
Underlying graph

Temporal paths: non strict and strict settings

Non-strict/Strict temporal path: A path and an **non-decreasing/increasing** sequence of times at which arcs can be traversed one after the other



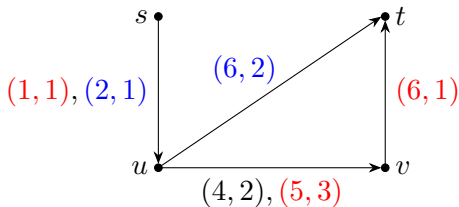
Non-strict setting



Strict setting

Our model

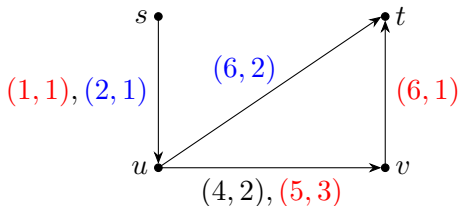
- Directed graph
- Delay: time it takes to traverse an arc



- Uniform delay zero = non-strict setting
- Uniform delay one = strict setting

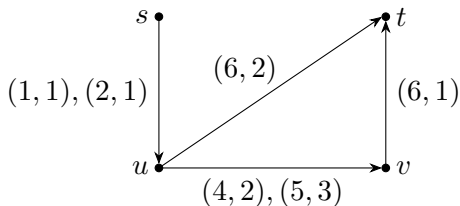
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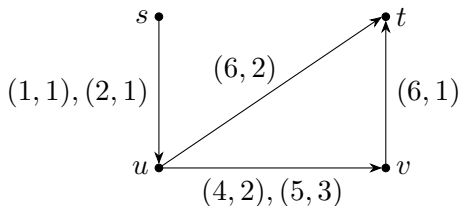
Our model



$$[(s, u, 1, 1), (s, u, 4, 2), (u, v, 4, 2) \dots]$$

- Each time we can traverse an arc $\Rightarrow$ a timed arc
- **Timed arc**: an arc, an appearance time, a delay (e.g. $(u, v, 4, 2)$ is a timed arc)
- Input: sorted list of timed arcs in non-decreasing order of appearance time
- M : number of timed arcs $\Rightarrow$ size of the input in $\mathcal{O}(M)$

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Generalization of static graph problems

- Revisit classical static graph problems in the context of temporal graphs
- Usually **more difficult**

Definition

A **temporal strongly connected component** (TSCC) is a maximal set of vertices U s.t. for any $s, t \in U$, there is a temporal path from s to t and from t to s using only timed arcs with extremities in U .

- With static graphs: find SCC in linear time
- With temporal graphs: find TSCC **NP-hard**

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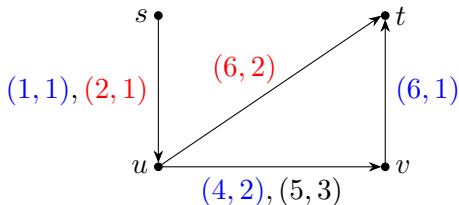
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In this talk: Restless Temporal Path

Δ -restless temporal path: Cannot wait more than Δ at an intermediate node



$$\Delta = 2$$

In this talk: Restless Temporal Path

RESTLESS-TEMPORAL-PATH

Input: A temporal graph G , a source node s , a target node t and a waiting time $\Delta \in \mathbb{N}$

Question: Is there a Δ -restless temporal path from s to t in G ?

Applications:

- Detection of a chain of infections in the SIR (Susceptible-Infected-Recovered) model
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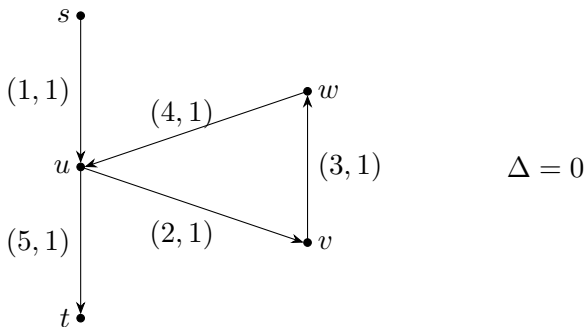
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In this talk: Restless Temporal Path

- Computing a restless temporal **walk**: **polynomial time** [Bentert et al. 2020]



- Computing a restless temporal **path**: **NP-hard** [Casteigts et al. 2021]

Parameters

NP-hard problems $\Rightarrow$ find parameters that **bring tractability**

Restless temporal path:

- Output-related parameter: the number of hops of the temporal path [Casteigts et al. 2021, Zschoche 2023]
- Classical parameter applied to the underlying graph: the feedback edge number of the underlying graph [Casteigts et al. 2021]
- New 'temporal' parameter: the timed feedback vertex number [Casteigts et al. 2021]

Invariant by permutation of the snapshots $\Rightarrow$ does not seem to really grasp the **temporal aspect**

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The (vertex)-interval-membership-width parameter

Interval-membership-width parameters: [Bumpus and Meeks 2022]

- Novel parameters that capture some temporal aspects
- Promising for reachability problems
 - FPT for temporal Euler circuit
 - NP-hard for MinReachDelete but FPT with the vertex variant
 - Counting the number of temporal paths [Enright et al. 2023]
 - Temporal Firefighter Problem [Hand et al. 2022]
- Incomparable with the other FPT parameters for the restless temporal path problem

Goal: Study this new parameter for the restless temporal path problem

The (vertex)-interval-membership-width parameter

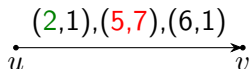
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Active window of an arc:



[first departure time of (u, v) , last arrival time of $(u, v)] = [2, 12]$

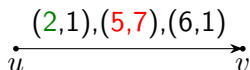
Active window of a vertex: Smallest window containing all active windows of arcs adjacent to the vertex



(vertex)-interval-membership-width: Maximum number of arcs (vertices) active at the same time

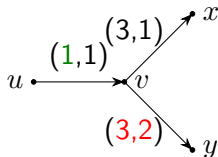
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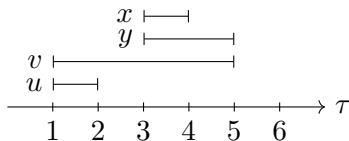


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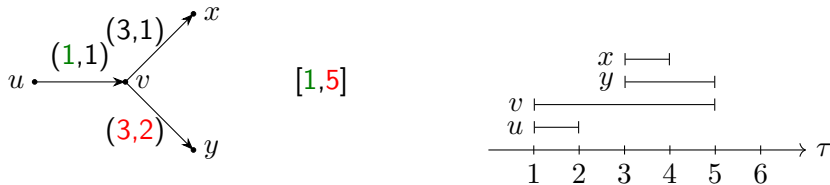
$[1, 5]$



(vertex)-interval-membership-width: Maximum number of arcs (vertices) active at the same time

A number line segment from u to v . Above the line, the points $(2,1)$, $(5,7)$, and $(6,1)$ are marked. The numbers 2 and 5 are red, and 1 and 7 are green.

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With the interval-membership-width

Proposition

RESTLESS-TEMPORAL-PATH is **NP-hard**, even if the input temporal graph has **interval-membership-width** equal to three and uniform delay one.

With the vertex-interval-membership-width

Theorem

When the input temporal graph $G = (V, E)$ has uniform delay one, RESTLESS-TEMPORAL-PATH can be solved in $\mathcal{O}(k2^k M)$ deterministic time, where k is the **vertex-interval-membership-width** of G .

General idea:

- Scan timed arcs in order of appearance time
- Maintain for each node u the Δ -restless temporal su -paths
- Problem: potentially **exponential** number
- Solution: store only traces of paths (only the nodes on the path that are active at considered time): at most **2^k traces**
- For each trace, store the **last arrival time** of a Δ -restless temporal path

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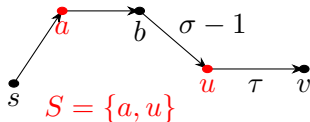
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Details: Scanning timed arcs at time τ :

- Basic step: Extending a found Δ -restless temporal path with timed arc (u, v, τ) :
 - Look at the couples (S, σ) in $L[u]$: $S \subseteq V$ is a trace, σ the last arrival time of an associated path
 - Check:
 - $v \notin S \Rightarrow$ valid temporal path
 - $\tau - \sigma \leq \Delta \Rightarrow \Delta$ -restless
 - Add $(S \cup \{v\}, \tau + 1)$ to $L'[v]$
- Clean up step:
 - Clean up lists by updating the traces and keeping the last arrival time for each trace



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With the vertex-interval-membership-width

- Previous algorithm cannot be easily adapted
- Reason: Implicitly assumes that all timed arcs with appearance time less than τ have arrival time at most τ
- Solution: store **all possible arrival times** at a node for each trace in a self-balancing binary search tree

Theorem

RESTLESS-TEMPORAL-PATH can be solved in $\mathcal{O}(k2^k M(k + \log M))$ deterministic time, where k is the **vertex-interval-membership-width** of G .

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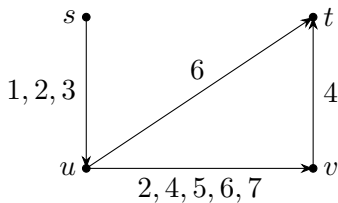
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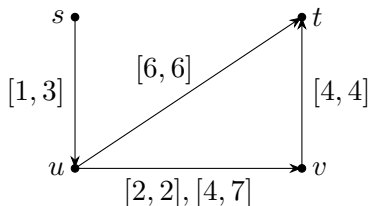
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Temporal graphs: interval model



Point model



Interval model

Interval model: More compact $\Rightarrow$ more difficult

NP-hardness

Theorem

In the interval model, RESTLESS-TEMPORAL-PATH is **NP-hard**, even if the input temporal graph has **vertex-interval-membership-width** equal to three.

- Reduction from SUBSET-SUM
- Uses intervals of exponential length

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Summary and future work

Contributions:

- **FPT algorithms** for the restless temporal path problem with the **vertex-interval-membership-width** in:
 - the point model with uniform delay one
 - the point model with arbitrary (positive) delays (additional computational cost)
- **NP-hardness** in:
 - the point model even if the interval-membership-width is equal to three
 - the interval model even if the vertex-interval-membership-width is equal to three

Open questions:

- Interval model with uniform delay one
- What other temporal parameters might bring tractability to the problem and are small in practice?

Thank you for listening!

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